UCI Control Target

Command Summary

Gideon Zweijtzer

All work Copyright © 2026 by
Gideon's Logic B.V.
All rights reserved.

Version 1.0, March 14, 2026

Table of Contents

1. Introduction **3**
1.1. Context 3
1.2. Purpose of this document 3
2. Commands **4**
2.1. CTRL_CMD_IDENTIFY (0x01) 4
2.2. CTRL_CMD_FINISH_CAPTURE (0x03) 4
2.3. CTRL_CMD_FREEZE (0x05) 4
2.4. CTRL_CMD_REBOOT (0x06) 4
2.5. CTRL_CMD_LOAD_REU (0x08) / SAVE_REU (0x09) 5
2.6. CTRL_CMD_U64_SAVEMEM (0x0F) 5
2.7. CTRL_CMD_DECODE_TRACK (0x11) 5
2.8. CTRL_CMD_GET_HWINFO (0x28) - DEPRECATED 6
2.9. CTRL_CMD_GET_DRVINFO ($29) 6
2.10. CTRL_CMD_ENABLE_DRIVE_A (0x30) 6
2.11. CTRL_CMD_DISABLE_DRIVE_A (0x31) 6
2.12. CTRL_CMD_ENABLE_DRIVE_B (0x32) 7
2.13. CTRL_CMD_DISABLE_DRIVE_B (0x33) 7
2.14. CTRL_CMD_GET_DRIVE_A_POWER (0x34) 7
2.15. CTRL_CMD_GET_DRIVE_B_POWER (0x35) 7
2.16. CTRL_CMD_GET_MP3_RAMDISKINFO (0x40) 7

1. Introduction

1.1. Context

The **"UCI Control Target"** provides a low-level interface to manage the hardware state of the Ultimate cartridge or Ultimate 64 machines and its emulated environment. This includes direct control over C64 execution (freeze/reboot), power management for emulated disk drives, and advanced utilities for REU and disk image handling.

The "UCI Control Target" is a target of the "Ultimate Command Interface" (UCI), and is thus accessible from the cartridge port, through some I/O registers. The document "Ultimate Command Interface – Register API" describes how commands are sent over this interface.

1.2. Purpose of this document

This document describes the commands that can be sent to this target, their expected parameters, and the data returned to the C64. These commands allow programs to interact with the Ultimate's management engine without leaving the C64 environment.

2. Commands

In the Ultimate products and the Commodore 64 Ultimate, the "Control Target" is accessible through target $04. This shall be the first byte of the command.

The following paragraphs describe each of the commands of "Control Target".

2.1. CTRL_CMD_IDENTIFY (0x01)

Command format: `$04 $01`

The "Identify" command sends back an identification string, such as: `"CONTROL TARGET V1.1"`. The user software can use this function to query which targets exist, or to obtain version information.

The status channel will report `00,OK`, as this command cannot fail.

2.2. CTRL_CMD_FINISH_CAPTURE (0x03)

Command format: `$04 $03`

Description: Finalizes an active tape capture session and closes the associated file.

Status: `00,OK`.

2.3. CTRL_CMD_FREEZE (0x05)

Command format: `$04 $05`

Description: Triggers a hardware freeze, equivalent to pressing the physical button on the Ultimate cartridge.

Status: `00, OK`

Note that the command still needs to be acknowledged after the freeze has completed. This might be problematic when the user selects something else in the menu. To be fixed / addressed.

2.4. CTRL_CMD_REBOOT (0x06)

Command format: `$04 $06`

Description: Triggers a full system reboot the C64. It corresponds to the menu item "Reboot" in the actions menu. This command also re-initializes the utility cartridge settings to the stored configuration.

Response: None

Status: None

Because the C64 gets reset; the Ultimate Command Interface (UCI) gets reset as well. Therefore no response is to be expected.

2.5. CTRL_CMD_LOAD_REU (0x08) / SAVE_REU (0x09)

Command format: `$04 $08 <FILENAME>` or `$04 $09 <FILENAME>`

Description: Loads an REU image from storage into the Ultimate's REU memory, or saves the current REU contents to a file.

Response: A 4-byte little-endian value representing number of bytes transferred. Negative values indicate an error; see status reply.

Status:

- `00,OK`
- `81,INVALID PARAMS`
- `84,REU NOT ENABLED`
- `85,REU FILE CANNOT BE OPENED.`

2.6. CTRL_CMD_U64_SAVEMEM (0x0F)

Command format: `$04 $0F <FILENAME>`

This command is only valid on the Ultimate 64, Ultimate 64 Elite, Ultimate 64 Elite-II and the Commodore 64 Ultimate.

This command saves the entire RAM. It does not save any other state information. When the filename is omitted, it will save by default to "`/temp/c64_memory.bin`"

Status:

- `00,OK`
- `87,DISK ERR: <error string>`

2.7. CTRL_CMD_DECODE_TRACK (0x11)

Command format: `$04 $11 <TRK> <MAX_SEC> <GCR_ADDR> <BIN_ADDR> <GCR_LEN>`

Description: High-speed GCR to Binary sector conversion.

Parameters:

- TRK is the expected track number.
- MAX_SEC is the highest sector number expected.
- GCR_ADDR is the offset in REU memory where the GCR data is located, LSB first.
- BIN_ADDR is the offset in REU memory where the decoded binary data will be stored.
- GCR_LEN is the length of the offered GCR track in bytes, LSB first.

Response: 1 byte (actual sectors) followed by 2 bytes of error flags per sector.

Status:

- `00,OK`
- `82,ERRORS ON TRACK`.

2.8. CTRL_CMD_GET_HWINFO (0x28) - DEPRECATED

Command format: `$04 $28 <SUB_CMD>`

Description: Queries hardware information. This command has two sub-commands, which violates the UCI concept. From firmware 3.15, when the sub command is not given, it defaults to returning the hardware model name in ASCII format.

- **Sub Command $00**: Returns the hardware model name (e.g., "ULTIMATE 64").
- **Sub Command $01:** Returns SID configuration, including base addresses and enable masks:
  - First byte contains the number of SID info frames follow.
  - Each SID info frame consists of 5 bytes:
    - Primary base address (2 bytes, LSB first)
    - Secondary base address (2 bytes, MSB first)
    - Type indicator, but unclear what it means; it seems not to match the actual implementation of SID control.

2.9. CTRL_CMD_GET_DRVINFO ($29)

Command format: `$04 $29 <EFFECTIVE_ADDR_FLAG>`

Description: This command returns the devices that the Ultimate

Response: Byte 0 is drive count, followed by 3-byte groups for each drive: `[Type]` `[IEC Bus Address]` `[Power State]`. The drive types are as follows:

Type	Value
1541	$00
1571	$01
1581	$02
Undecided	$03
Software IEC	$0F
Printer	$50

The power state is either `$00` or `$01`.

Status: `00,OK`

2.10. CTRL_CMD_ENABLE_DRIVE_A (0x30)

Command format: `$04 $30`

Description: Controls the power state of emulated drive A: This command turns the drive ON.

Status: `00,OK`

2.11. CTRL_CMD_DISABLE_DRIVE_A (0x31)

Command format: `$04 $31`

Description: Controls the power state of emulated drive A: This command turns the drive OFF.

Status: `00,OK`

2.12. CTRL_CMD_ENABLE_DRIVE_B (0x32)

Command format: `$04 $32`

Description: Controls the power state of emulated drive B: This command turns the drive ON.

Status: `00,OK`

2.13. CTRL_CMD_DISABLE_DRIVE_B (0x33)

Command format: `$04 $33`

Description: Controls the power state of emulated drive B: This command turns the drive OFF.

Status: `00,OK`

2.14. CTRL_CMD_GET_DRIVE_A_POWER (0x34)

Command format: `$04 $34`

Description: With this command the power state of drive A can be queried.

Response: "`off`" **or** "`on  `"

Status: `00,OK`

2.15. CTRL_CMD_GET_DRIVE_B_POWER (0x35)

Command format: `$04 $35`

Description: With this command the power state of drive B can be queried.

Response: "`off`" **or** "`on  `"

Status: `00,OK`

2.16. CTRL_CMD_GET_MP3_RAMDISKINFO (0x40)

Command format: `$04 $40`

Description: This is a command specifically made for Geos MegaPatch 3. It returns the configuration of the RAM disks found in the REU image.

Response: An 8-byte status block (2 bytes per drive) indicating RAM disk types (1541, 1571, 1581, or Native) and their sizes. The first of the two bytes indicates the type (0x41, 0x71, 0x81 or 0xDD). For native partitions, the second byte indicates the size expressed in blocks of 64 KiB, otherwise it's zero.

Status: `00,OK`